India- Industries -Geographical Perspective

The Eight Core Industries connect Indian industrial geography, mineral resources, energy, infrastructure, regional development and economic growth.

Let's discuss these through Three lenses

1. Geographical location — Why is the industry located there? Raw materials, energy, water, transport, ports, markets and historical factors.

2. Economic importance — Its contribution to infrastructure, manufacturing, employment, investment and supply chains.

3. Andhra Pradesh and Telangana — Important industrial centres, resource linkages, regional advantages and UPSC-style geographical reasoning.

One foundational distinction: The Eight Core Industries are an official economic indicator group, not a list of eight industries that are necessarily the largest employers or the largest contributors to GDP.

Their combined performance is tracked through the Index of Eight Core Industries (ICI), which is an important lead indicator of industrial activity.

India's Industrial Geography • Economy • Resources • Andhra Pradesh & Telangana

1. First, understand the change: Eight Core Industries → Nine-Core Series

This is an important current-affairs update . Traditionally, India has tracked eight core industries. In the revised series with 2022–23 as the base year, iron ore has been added as the ninth component. Therefore, students must distinguish the conventional Eight Core Industries from the revised nine-component index.

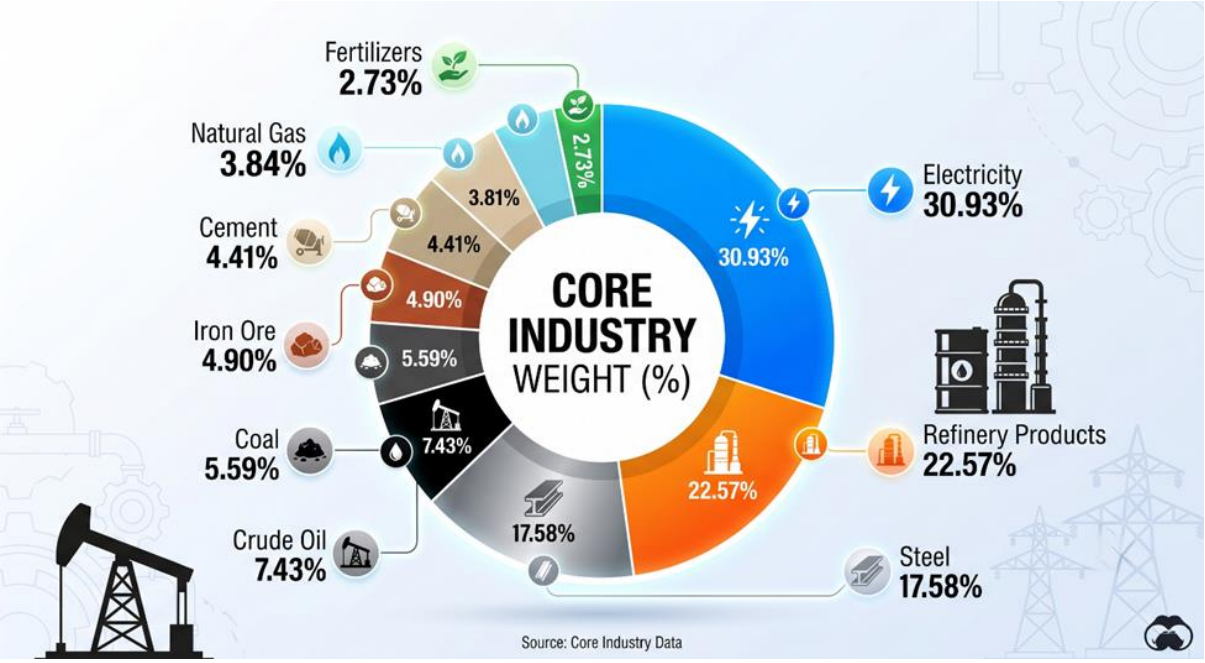
Industry	Weight in revised index (%)
Electricity	30.932
Petroleum refinery products	22.572
Steel	17.584
Crude oil	7.430
Coal	5.596
Iron ore	4.905

Cement	4.410
Natural gas	3.841
Fertilizers	2.731
Total	100.000

Source: Revised ICI weights based on the 2022–23 IIP series. Figures are rounded. And CMIE (Centre for Monitoring Indian Economy)

Relative weight of the nine components

Percentage weights in the revised Index of Core Industries, base year 2022–23.



What should we infer from these weights?

- Electricity has the largest weight, reflecting the importance of power generation to industrial activity.

- Petroleum refining and steel together have a substantial influence on the index.

- Fertilizers have a relatively small weight, but they remain strategically important for agricultural productivity and food security.

- Weight is not the same as growth rate, physical output, GDP contribution or employment share.

- A rise in the index does not necessarily mean that every component has grown.

The index is compiled and released monthly by the Office of the Economic Adviser under the Department for Promotion of Industry and Internal Trade (DPIIT). It provides an early indication of industrial performance ahead of the broader Index of Industrial Production (IIP). Indian Economic Service+CMIE

2. Industrial location: the geographical foundation

Why is an industry located at a particular place?

Raw materials

Coal, iron ore, limestone, crude oil, natural gas and agricultural inputs.

Energy and water

Reliable electricity, cooling water, process water and fuel.

Transport and ports

Railways, highways, pipelines, ports and freight terminals.

Markets and labour

Demand, skilled workers, technology, finance and supplier networks.

The key geographical principle: industries differ in their sensitivity to transport costs, raw materials, energy, markets and agglomeration economies.

Three classic location patterns help to organise this topic:

- Raw-material-oriented: coal mining, iron ore mining and many cement plants.

- Port- or coast-oriented: oil refineries, petrochemicals and some import-dependent industrial complexes.

- Market-, infrastructure- or network-oriented: many fertilizer plants, electricity-intensive manufacturing units and downstream steel industries.

These are tendencies, not absolute rules. Modern pipelines, rail networks, imported raw materials and industrial corridors can change traditional location patterns.

3. The nine industries: geography and economic importance

1. Coal- Geographical logic - Coal is concentrated mainly in the Gondwana sedimentary basins of eastern and central India. Important coal-bearing regions

include the Damodar Valley, Mahanadi Valley, Son Valley, Godavari Valley and parts of the Wardha Valley.

Major Indian locations

- Jharkhand: Jharia, Bokaro and North Karanpura.

- Odisha: Talcher and Ib Valley.

- Chhattisgarh: Korba and Mand–Raigarh.

- Madhya Pradesh–Uttar Pradesh region: Singrauli.

- Telangana: Godavari Valley coalfield, including the Singareni mining belt.

Economic importance

Coal supports thermal electricity, steelmaking, cement, industrial heating and rail freight. Its location influences power-plant siting, industrial towns and regional employment, while also raising concerns about air pollution, land degradation and mine displacement.

Andhra Pradesh and Telangana connection

- Telangana: Singareni Collieries operates in the Godavari Valley coal belt. Ramagundam and Godavarikhani are important coal-and-energy centres.

- Andhra Pradesh: Coal availability is more limited; power generation depends on a combination of imported and domestic coal, other fuels and renewable electricity. (The Ministry of Coal lists Telangana among India's major coal-resource states). Ministry of Coa

UPSC geographical trap: Coal deposits and coal consumption are not identical spatial patterns. A state can generate coal-based electricity using coal transported from another state or imported through a port.

2. Crude oil- Geographical logic- Petroleum occurs mainly in sedimentary basins with suitable source rocks, reservoir rocks and geological traps. Both onshore and offshore exploration are important.

Major Indian locations

- Assam–Arakan Basin: Digboi, Naharkatiya and Moran areas.

- Mumbai Offshore: Mumbai High and adjoining offshore fields.

- Cambay Basin: Gujarat.

- Rajasthan: Barmer basin.

- Krishna–Godavari Basin: onshore and offshore areas of Andhra Pradesh and the adjoining Bay of Bengal.

Economic importance

Crude oil is the primary feedstock for refineries and contributes to transport fuels, petrochemicals and industrial energy. Domestic production reduces import exposure, although India continues to depend substantially on imported crude oil.

AP and Telangana connection

- Andhra Pradesh: The Krishna–Godavari Basin is the principal regional example. It contains important hydrocarbon-bearing onshore and offshore areas.

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- Telangana: It is not a major producing crude-oil region comparable to the main petroleum basins. Its fuel requirements are linked to interstate transport, refineries and distribution networks.

UPSC geographical trap: The presence of a sedimentary basin does not guarantee commercially recoverable oil in every part of that basin.

3. Natural gas- Geographical logic

Gas is found in petroleum-bearing sedimentary basins and in gas-specific reservoirs. Its movement depends heavily on pipelines, LNG terminals and processing facilities.

Major Indian locations

- Krishna–Godavari offshore basin.

- Assam and Gujarat gas-producing regions.

- Western offshore fields, including the Mumbai Offshore region.

- LNG import and regasification facilities on the coast connect imported gas to inland consumers.

Economic importance- Natural gas supplies fertilizer plants, city-gas networks, industrial boilers, power plants and petrochemical industries. It is particularly important where plants can obtain a reliable pipeline supply.

AP and Telangana connection

- Andhra Pradesh: The Krishna–Godavari Basin and coastal gas infrastructure are strategically important. The Kakinada–Srikakulam gas pipeline project illustrates how a pipeline can connect coastal districts to a wider gas network.

National Data Repository+1

- Telangana: Ramagundam Fertilizers and Chemicals Limited (RFCL) illustrates the relationship between gas supply and fertilizer production. Hyderabad and other urban-industrial centres also depend on gas distribution networks.

UPSC geographical trap: A gas-producing region and a gas-consuming industrial region need not coincide; pipelines connect the two.

4. Petroleum refinery product- Geographical logic

Refineries are often located near ports, crude-oil fields, pipeline terminals, major markets or combinations of these factors. Refineries convert crude oil into usable fuels and feedstocks.

Major Indian locations

- Western coast: Jamnagar, Vadinar and Mumbai.

- Eastern coast: Visakhapatnam, Paradip and Haldia.

- Southern India: Chennai and Kochi.

- Inland: Mathura, Panipat, Barauni and Bina, among others.

Economic importance

Refining supports petrol, diesel, aviation turbine fuel, LPG, lubricants and petrochemical feedstocks. It creates demand for storage, shipping, pipelines, chemicals, engineering and downstream manufacturing.

AP and Telangana connection

- Andhra Pradesh: HPCL's Visakhapatnam refinery is a major industrial anchor. ONGC's Tatipaka installation is a much smaller refining facility associated with offshore/onshore hydrocarbon operations. (Government of India)

- Telangana: It has no comparable large crude-oil refinery; fuel reaches the state through transport and distribution networks from refineries elsewhere.

Why Visakhapatnam? Its coastal location facilitates crude-oil imports, marine logistics and the distribution of refined products. Its industrial ecosystem also supports engineering, shipping and downstream activities.

5. Fertilizers- Geographical logic- Fertilizer location depends on feedstock availability, energy, water, transport and proximity to agricultural markets. Gas is a major feedstock for modern nitrogenous fertilizer plants; phosphate and potash supply chains depend more heavily on mineral resources and imports.

Major Indian locations

- Gujarat: gas-based fertilizer and chemical complexes.

- Uttar Pradesh and Bihar: fertilizer plants serving major agricultural regions.

- Odisha: fertilizer and chemical facilities supported by coastal and mineral linkages.

- Punjab and Haryana: plants and distribution networks linked to intensive agricultural demand.

- Andhra Pradesh and Telangana: coastal chemical-industrial infrastructure and regional fertilizer production.

Economic importance

Fertilizers replenish essential plant nutrients, improve crop productivity and support food security. Their location affects the cost and reliability of agricultural input distribution.

AP and Telangana connection

- Andhra Pradesh: Visakhapatnam hosts Coromandel International's fertilizer operations and has a broader chemical-industrial base. Coastal access supports the handling of imported raw materials.

- Telangana: RFCL at Ramagundam is a prominent urea producer. Its location illustrates the relationship between gas availability, industrial infrastructure and fertilizer demand.

UPSC geographical trap: A state can have large fertilizer consumption without producing all the fertilizers it uses. Consumption follows agricultural demand, while production follows feedstock and industrial economics.

6. Steel- Geographical logic- Integrated steel plants traditionally favour access to iron ore, coking coal, limestone, water, power and rail transport. Coastal plants can use imported coking coal and serve export markets.

Major Indian locations

- Jharkhand: Jamshedpur and Bokaro.

- Odisha: Rourkela, Angul and Kalinganagar.

- Chhattisgarh: Bhilai.

- West Bengal: Durgapur and Burnpur.

- Karnataka: Vijayanagar region.

- Andhra Pradesh: Visakhapatnam.

Economic importance

Steel is essential for infrastructure, railways, automobiles, machinery, defence production, construction and capital goods. Steel plants generate demand for mining, transport, power, fabrication and engineering services.

AP and Telangana connection

- Andhra Pradesh: Visakhapatnam Steel Plant is the state's most prominent integrated steelmaking centre. Its coastal location helps with imported raw materials and maritime logistics.

- Telangana: It has mineral resources and metal-processing activities, but lacks a steel complex of the scale of Visakhapatnam's integrated plant. Bayyaram's iron-ore deposits are a useful case study in the gap between mineral occurrence and the establishment of a commercially viable steel plant.

UPSC geographical trap: Iron ore alone cannot determine steel-plant location. Coking coal, water, transport, power, technology and market access also matter.

7. Cement- Geographical logic- Cement plants tend to locate near limestone deposits because limestone is the principal raw material and transporting it over long distances can be costly. Coal or other fuels, electricity, gypsum and transport also matter.

Major Indian locations

- Rajasthan: extensive limestone-bearing regions.

- Madhya Pradesh and Chhattisgarh: limestone and energy linkages.

- Andhra Pradesh and Telangana: major limestone-bearing belts.

- Gujarat and Tamil Nadu: limestone, markets and transport advantages.

Economic importance

Cement is fundamental to housing, roads, bridges, dams, ports, industrial buildings and urban infrastructure. The industry links mining to construction and public capital expenditure.

AP and Telangana connection

- Andhra Pradesh: Kadapa, Kurnool, Nandyal and other limestone-bearing areas support cement production. The Rayalaseema region is especially relevant when teaching resource-based industry.

- Telangana: Suryapet, Nalgonda, Peddapalli, Mancherial and other limestone-bearing areas support cement and related mineral industries. Peddapalli district's industrial profile includes cement production.

UPSC geographical trap: Cement plants are often raw-material-oriented, but the distribution of cement markets, railway freight rates and infrastructure corridors also influences location.

8. Electricity- Geographical logic -Electricity generation is distributed across thermal, hydro, nuclear and renewable sources. Each has a different location logic: coal and water for thermal plants, river gradients and storage for hydropower, suitable geology and infrastructure for nuclear power, and solar or wind resources for renewables.

Major Indian locations

- Coal-based thermal generation: Odisha, Chhattisgarh, Jharkhand, Madhya Pradesh, Telangana and other coal-consuming regions.

- Hydropower: Himalayan rivers and the Western Ghats.

- Nuclear generation: selected coastal and inland sites.

- Solar power: Rajasthan, Gujarat and other high-insolation regions.

- Wind power: Tamil Nadu, Gujarat, Karnataka, Maharashtra and Rajasthan.

Economic importance

Electricity is a general-purpose input for almost every sector. It supports manufacturing, irrigation, transport, digital services, households and public infrastructure.

AP and Telangana connection

- Telangana: Ramagundam is a major thermal power centre. The state's electricity system also includes other thermal and renewable installations.

- Andhra Pradesh: Thermal generation, hydropower and solar and wind resources contribute to the power mix. The state's long coastline, river systems and inland renewable-energy potential create a diverse geographical setting.

UPSC geographical trap: Installed capacity is not the same as actual electricity generation. Generation depends on plant availability, fuel supply, water availability, demand and renewable-resource conditions.

9. Iron ore — the additional component- Geographical logic- Iron ore is associated with ancient crystalline rocks and banded iron formations. Mining is therefore concentrated in mineral-rich geological belts rather than being distributed according to population alone.

Major Indian locations

- Odisha: Keonjhar, Sundargarh and adjoining mineral belts.

- Jharkhand: Singhbhum region.

- Chhattisgarh: Bailadila and Dalli-Rajhara regions.

- Karnataka: Ballari–Hospet and adjoining belts.

- Goa and Maharashtra: additional deposits and mining regions.

Economic importance

Iron ore is a principal raw material for the iron and steel industry. Its quality, grade, beneficiation requirements, transport cost and access to steel plants affect its commercial value.

AP and Telangana connection

- Andhra Pradesh: Iron ore occurs in several districts, including parts of Anantapur, Kadapa, Kurnool and Prakasam.

- Telangana: Deposits occur in parts of the state, including the Bayyaram region and scattered occurrences in other districts. Their existence does not automatically establish economic viability for a large steel plant.

UPSC geographical trap: The addition of iron ore to the revised index does not mean that it has suddenly become a newly discovered mineral or a new industry. It means its production has been added as a separately tracked component of the revised index.

4. Andhra Pradesh vs Telangana: industrial geography at a glance

This is the section for APPSC, TGPSC and UPSC students.

Andhra Pradesh — the coastal industrial advantage

Key geographical advantages

- Long coastline, ports and maritime connectivity.

- Krishna–Godavari hydrocarbon basin.

- Limestone and other mineral resources, particularly in parts of Rayalaseema.

- Fertilizer, refinery, steel, port-based and agro-processing industries.

- Industrial-corridor development and multimodal logistics.

Important examples: Visakhapatnam — refining, steel and fertilizers; Krishna–Godavari Basin — oil and gas; Rayalaseema — limestone and cement; coastal industrial belt — ports, chemicals and logistics.

Telangana — the mineral-energy-industrial advantage

Key geographical advantages

- Godavari Valley coalfields and the Singareni mining belt.

- Thermal power and coal-linked industrial activity.

- Limestone resources supporting cement production.

- Hyderabad's skilled workforce, infrastructure, research institutions and industrial services.

- Engineering, pharmaceuticals, electronics and technology-based manufacturing alongside traditional resource-based industries.

Important examples: Ramagundam–Godavarikhani — coal and electricity; Ramagundam — fertilizer production; Nalgonda–Suryapet and adjoining regions — limestone and cement; Hyderabad–Rangareddy industrial region — diversified manufacturing and technology-based activities.

ramagundamcorporation.telangana.gov.in+3

5. Industry-location matrix for classroom revision

Industry Primary location factor Andhra Pradesh Telangana

Coal Geological deposits Limited compared with major coal belts Singareni–Godavari Valley

Crude oil Sedimentary basins Krishna–Godavari Basin Not a major producing region

Natural gas Gas fields and pipelines Krishna–Godavari Basin, coastal infrastructure Gas-based industrial consumption

Refinery products Ports, crude supply, markets Visakhapatnam refinery Dependent on supplies from refineries elsewhere

Fertilizers Gas, feedstock, agricultural demand Visakhapatnam Ramagundam

Steel Ore, coal, water, transport Visakhapatnam Mineral potential; no comparable large integrated plant

Cement Limestone, fuel, transport Kadapa–Kurnool–Nandyal belt Nalgonda–Suryapet and other limestone belts

Electricity Fuel and renewable resources Thermal, hydro and renewables Ramagundam and other generation centres

Iron ore Mineral-bearing geological formations Deposits in several districts Bayyaram and scattered deposits

6. Five high-value geographical insights for UPSC Prelims

Insight 1 — Mineral belts create industrial clusters

Coal, iron ore and limestone can support power, steel and cement clusters. However, the existence of resources alone is insufficient; transport, energy, water, investment and markets also determine viability.

Insight 2 — Coastal location can compensate for resource scarcity

Visakhapatnam can support a large refinery and steel complex despite not being located in the heart of India's principal coalfields. Ports allow imported raw materials to reach industrial users.

Insight 3 — Pipelines alter traditional industrial geography

Natural gas can connect offshore fields or LNG terminals to inland fertilizer plants and industrial consumers. This reduces dependence on physical proximity to the source.

Insight 4 — Industrial location and agricultural demand are linked

Fertilizer distribution depends on cropping intensity and nutrient demand, while fertilizer production depends on feedstock and plant economics. The two distributions need not coincide.

Insight 5 — Core industries are interdependent

Coal supports electricity; electricity supports steel and cement; steel and cement support infrastructure; petroleum and gas support transport, chemicals and fertilizers. A disruption in one sector can spread through the industrial system.

India's Industrial System

Resources • Energy • Infrastructure • Markets

Resource base

Coal • Iron ore • Limestone • Oil and gas

Energy system

Electricity • Fuels • Gas networks

Core industrial activities

Mining • Refining • Steel • Cement • Fertilizers

Economic outcomes

Infrastructure • Employment • Manufacturing • Exports • Regional development